Rubber-Band Rover Energy Budget

Build and test a rubber-band-powered vehicle, then account for where stored energy goes.

Intermediate	1 class period	Skill Builder

Engineering Context

A lightweight inspection rover needs a simple stored-energy demonstration for a training program. The vehicle must travel predictably rather than merely as far as possible.

What You Will Practice

- Describe energy transformations
- Measure distance and time
- Control experimental variables
- Use data to select an operating condition

What You Need

- Small wheeled chassis materials
- Rubber bands
- Meter tape
- Stopwatch
- Masses
- Engineering notebook

Student Instructions

1. Build a vehicle with a repeatable rubber-band winding method.
2. Choose three winding levels such as 10, 20, and 30 turns.
3. Run three trials at each level.
4. Record distance, travel time, and any wheel slip or path deviation.
5. Calculate average distance and average speed.
6. Create a graph of winding level versus distance.
7. Select the best operating condition for predictable travel and justify it.

Engineering Requirements

- Same vehicle configuration for all baseline trials
- Three trials per winding level
- Release method must be consistent
- Vehicle must remain within a defined lane
- Recommendation must use data, not the single longest run

Constraints & Safety

- Wear eye protection if required
- Inspect rubber bands for cracks
- Keep faces and hands away from stretched bands
- Use a clear test lane

What You Submit

- Vehicle sketch
- Trial data table
- Graph
- Energy-flow diagram
- Operating recommendation

Reflection Questions

1. What evidence suggested energy was lost to friction or wheel slip?
2. Why did the highest winding level not necessarily give the best result?
3. How did mass affect acceleration or consistency?
4. What variable would you test next?

Engineering Tip Mark the axle so every trial begins with the same winding reference.	**Common Mistake** Changing the release angle or starting position between trials makes the data difficult to compare.
Stretch Goal Add payload and determine the best winding level for transporting it.	**Career Connection** Test engineers characterize stored-energy devices, springs, deployment systems, and small mobility platforms before mission use.